

DEPARTMENT OF Information Technology
OPERATING SYSTEMS (IT-4005)
Special ASSIGNMENT

- Q 1. Critically analyze the Reader–Writer Problem in the context of process synchronization.**
- Q 2. Examine the Critical Section Problem and evaluate multiple synchronization mechanisms for its resolution.**
- Q 3. Describe the disk architecture and critically evaluate a minimum of five disk scheduling algorithms**
- Q 4. Elaborate on the concepts of Protection, Security, and Threats in the context of a File Management System.**
- Q 5. Discuss an Operating System and provide a comprehensive classification of OS types based on their structure, functionality, and application domain.**
- Q 6. Explain Process with elaborating on the Process State Transition Diagram and discuss CPU scheduling criteria and algorithms comprehensively.**
- Q 7. Solve the following scheduling numerical completely, showing all intermediate steps and computing all performance metrics. Calculate Gantt chart, TAT, CT, and their average with CPU burst time.**

Part A — Round Robin Scheduling (Time Quantum = 3 units)

Process	Arrival Time	Burst Time
P1	0	10
P2	2	6
P3	4	14
P4	6	8
P5	8	4

Part B — Highest Response Ratio Next (HRRN) Scheduling

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	9

P4	5	3
P5	8	5
P6	10	2

Q 8. Apply the Banker's Algorithm to determine safe states and handle resource requests : A system has three resource types: A, B, C. The total resources in the system are: A = 10, B = 5, C = 7. At time T_0 , the current allocation and maximum resource requirement for each process are given below:

Allocation and Maximum Table:

Process	Alloc A	Alloc B	Alloc C	Max A	Max B	Max C
P0	0	1	0	7	5	3
P1	2	0	0	3	2	2
P2	3	0	2	9	0	2
P3	2	1	1	2	2	2
P4	0	0	2	4	3	3

Available Resources at T_0 :

Resource A	Resource B	Resource C
3	3	2

Q 9. Discuss semaphore-based synchronization through theoretical exposition and numerical simulation.

Q 10. Apply and critically evaluate multiple page replacement algorithms on a given reference string: Consider a process with a reference string of 15 page references and a physical memory allocation of 3 frames (initially all empty). Apply each page replacement algorithm to the complete reference string.

Reference String: 3, 5, 2, 1, 3, 4, 2, 5, 3, 1, 2, 4, 3, 5, 1

Q 11. Solve a comprehensive set of numerical problems on the Paging memory management scheme.

A system uses 32-bit logical addressing with a page size of 4 KB. Each page table entry occupies 4 bytes.

(a) Fundamental Calculations: Compute (i) the number of pages in the logical address space, (ii) the number of bits used for the page number and the page offset in a logical address, (iii) the size of a single-level page table for one process.

(b) Logical to Physical Address Translation: Given the following page table for a process (page size = 512 bytes = 2^9 , so 9-bit offset), translate the following logical addresses into physical addresses. Show the decomposition of each logical address into (Page Number, Offset) and compute the Physical Address = Frame Number $\times$ Page Size + Offset:

Page Number	Frame Number	Offset	Physical Address
0	5	—	—
1	10	—	—
2	3	—	—
3	18	—	—
4	7	—	—

Translate the following logical addresses (expressed in decimal):

(i) 1052 | (ii) 2500 | (iii) 0 | (iv) 4095 | (v) 3100

(c) Two-Level Paging: For the same 32-bit system, if two-level paging is used with 10 bits for the outer page number, 10 bits for the inner page number, and 12 bits for the offset, compute the size of the outer page table and the maximum number of inner page tables possible.

(d) Inverted Page Table: Explain the structure of an Inverted Page Table. For a system with 2 GB physical memory and 4 KB page size, compute the number of entries in the Inverted Page Table and compare its size (in bytes, at 8 bytes/entry) with a conventional page table for a process with a 4 GB virtual address space at 4 bytes/entry.

(e) Internal Fragmentation Analysis: A process requires exactly 13,500 bytes of memory. If the page size is 4096 bytes, how many pages does the process occupy? How much internal fragmentation occurs? What percentage of the last page is wasted?

(f) TLB Performance: A system has a TLB hit ratio of 90%. Memory access time is 100 ns. TLB access time is 10 ns. Compute the Effective Memory Access Time (EMAT) with and without TLB. What TLB hit ratio is required to achieve an EMAT within 10% of main memory access time?